

exp0526: Pure 0522-Style Closed-Loop Controller Learning for Age-Structured Predator–Prey PDEs

exp0526 notes

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1 Purpose

This note reformulates the age-structured predator–prey control setting from *Lotka–Sharpe Neural Operators for Control of Population PDEs* as a pure exp0522-style closed-loop controller learning problem [1].

The first exp0526 version should not learn a Lotka–Sharpe neural operator and should not approximate the paper’s analytic controller. Instead, the model directly outputs a scalar control signal $u(t)$, applies it to the population dynamics, and receives loss from the resulting distance to the equilibrium state.

The controller interface is

$$(\eta_1(t), \eta_2(t), 1; \text{ internal self-talk}) \mapsto u(t).$$

The ordinary external observation is (η_1, η_2) together with a constant pulse 1. The self-talk state is internal to the exp0522 model: the previous language output is fed into the next step.

2 Problem From the Miroslav Paper

The environment is the two-species age-structured predator–prey PDE used in the Lotka–Sharpe neural-operator paper and inherited from the age-structured predator–prey stabilization literature [1, 4]. It is a McKendrick–von Foerster type age-structured model [2]:

$$\partial_t x_1(a, t) + \partial_a x_1(a, t) = -x_1(a, t) \left[\mu_1(a, t) + u(t) + \int_0^A g_1(\alpha) x_2(\alpha, t) d\alpha \right], \quad (1a)$$

$$\partial_t x_2(a, t) + \partial_a x_2(a, t) = -x_2(a, t) \left[\mu_2(a, t) + u(t) + \frac{1}{\int_0^A g_2(\alpha) x_1(\alpha, t) d\alpha} \right], \quad (1b)$$

$$x_i(0, t) = \int_0^A k_i(a, t) x_i(a, t) da, \quad i = 1, 2. \quad (1c)$$

Here $x_i(a, t) > 0$ are the age profiles, k_i are fertility profiles, μ_i are mortality profiles, g_i are interaction kernels, and $u(t) \geq 0$ is the scalar dilution or harvesting input.

The control target is a positive equilibrium profile

$$x_i(a, t) \rightarrow x_i^*(a, t), \quad i = 1, 2,$$

or, equivalently for the reduced controller coordinates, the equilibrium

$$\eta_1(t) \rightarrow 0, \quad \eta_2(t) \rightarrow 0.$$

3 Equilibrium Error Coordinates

For each species, the Lotka–Sharpe scalar is defined implicitly by the classical Lotka–Sharpe age-distribution condition [3]:

$$\int_0^A k_i(a, t) \exp \left(- \int_0^a (\mu_i(s, t) + \zeta_i(t)) \, ds \right) \, da = 1. \quad (2)$$

The Lotka–Sharpe neural-operator paper uses this quantity to construct transformed coordinates [1]

$$\eta(t) = (\eta_1(t), \eta_2(t)).$$

In the paper’s notation,

$$\eta_i(t) = \log \left(\frac{\langle \pi_{0,i}(\cdot, t), x_i(\cdot, t) \rangle}{x_i^*(0, t) \kappa_i(t)} \right). \quad (3)$$

The exp0526 controller observes η_1, η_2 , not the full hidden ecological parameter state. The full PDE remains the environment, while η is the low-dimensional control error exposed to the agent.

4 What We Change

The Lotka–Sharpe neural-operator paper keeps the analytic feedback law and uses learning only to approximate the implicit Lotka–Sharpe map

$$(k, \mu) \mapsto \zeta.$$

That is not the first exp0526 task.

By contrast, the first exp0526 task is direct closed-loop controller learning:

$$(\eta_1(t), \eta_2(t), 1; m_{t-1}) \mapsto u(t),$$

where m_{t-1} is the model’s own previous language output. The model is judged by what its control does to the environment, not by whether it matches an analytic control formula.

Problem 1 (Pure exp0526 closed-loop controller). *Train an exp0522-style self-talk MLP inside the full two-species age-structured PDE environment (1). At each time step the agent observes $\eta_1(t)$, $\eta_2(t)$, and a constant pulse 1, produces a scalar control $u(t)$, and is optimized through the resulting future equilibrium error.*

5 0522 Mechanism Reused

The exp0522 mechanism is not an external memory module. It is a recurrent self-talk loop created by feeding the previous language output into the next MLP step.

At step t , the hidden network computes

$$h_t = F_\theta(\eta_1(t), \eta_2(t), 1, e_{t-1}, m_{t-1}),$$

then emits both a control preactivation and a new language state:

$$\tilde{u}_t = W_u h_t + b_u, \quad (4)$$

$$m_t = R h_t. \quad (5)$$

Here R is the fixed sparse signed readout used by exp0522. The next step receives m_t . This is the sense in which the model has memory: the language output from the previous step becomes part of the next step’s input.

The ordinary problem statement remains

$$(\eta_1, \eta_2, 1) \mapsto u,$$

because m_t is an internal model state rather than an externally measured ecological variable.

Error Channel

In exp0522 prediction, the error channel is a scalar prediction error. In exp0526 control, the first version should reinterpret the error channel as a control-error view:

$$e_t = \eta_t = (\eta_1(t), \eta_2(t)).$$

This is not a control-label error. It is the current distance direction from equilibrium, made available through the same error-feedback interface so that the 0522 ablations remain meaningful.

The no-error comparison forces this channel to zero. The no-language comparison uses $m_t \equiv 0$. Blink and stutter tests temporarily remove the error or language pathway during rollout.

6 Closed-Loop Learning Objective

The network produces an unconstrained scalar $\tilde{u}_t$. The actual control applied to the environment is

$$u_t = \text{softplus}(\tilde{u}_t),$$

which enforces $u_t \geq 0$ while preserving smooth gradients.

For one differentiable rollout window of length T , the main objective is distance to equilibrium:

$$\mathcal{L}_\eta = \frac{1}{T} \sum_{t=1}^T (\eta_1(t)^2 + \eta_2(t)^2). \quad (6)$$

A small control-energy regularizer can be added:

$$\mathcal{L} = \mathcal{L}_\eta + \lambda_u \frac{1}{T} \sum_{t=1}^T u_t^2, \quad 0 < \lambda_u \ll 1. \quad (7)$$

The required gradient path is

$$\mathcal{L} \rightarrow \eta_{t+1} \rightarrow x(\cdot, t+1) \rightarrow u_t \rightarrow \tilde{u}_t \rightarrow h_t \rightarrow \theta.$$

Because $m_t = Rh_t$ is fed to later steps, gradients inside a training window also flow through the self-talk loop. At window boundaries, the message and PDE state are carried forward but detached, matching the continuous-window protocol from exp0522.

Backpropagation Through the Discrete Rollout

Question. How exactly does the equilibrium loss produce a gradient with respect to the controller parameters?

In implementation, the continuous PDE is replaced by a differentiable time-step map. For a fixed hidden ecological profile at step n , write the age-grid state as $X_n \in \mathbb{R}^{2 \times N_a}$ and the controller state as m_{n-1} . The model and environment form the closed-loop recursion

$$o_n = [\eta(X_n; \xi_n), 1], \quad (8)$$

$$(\tilde{u}_n, m_n) = C_\theta(o_n, e_n, m_{n-1}), \quad (9)$$

$$u_n = \text{softplus}(\tilde{u}_n), \quad (10)$$

$$X_{n+1} = \Phi_{\Delta t}(X_n, u_n; \xi_n), \quad (11)$$

where $\xi_n = (k_n, \mu_n, g_n, \zeta_n, \pi_{0,n}, x_n^*)$ denotes the environment quantities used at that step. The map $\Phi_{\Delta t}$ is the upwind age-grid update together with the renewal boundary condition. The training loss over one window $n = 0, \dots, T-1$ is

$$\mathcal{L}_T(\theta) = \frac{1}{T} \sum_{n=1}^T \|\eta(X_n; \xi_n)\|_2^2 + \lambda_u \frac{1}{T} \sum_{n=0}^{T-1} u_n^2. \quad (12)$$

Thus the controller is not trained from a target control label. It is trained by differentiating the consequence of its own control actions through the rollout.

Question. What chain rule is being used inside one training window?

Let $z_n = (X_n, m_{n-1})$ be the recurrent closed-loop state. Equations (11) define a differentiable recurrence

$$z_{n+1} = G_\theta(z_n; \xi_n).$$

Backpropagation through time computes

$$\nabla_\theta \mathcal{L}_T = \sum_{n=0}^{T-1} \left(\frac{\partial \ell_n}{\partial u_n} \frac{\partial u_n}{\partial \tilde{u}_n} \frac{\partial \tilde{u}_n}{\partial \theta} \right) + \sum_{n=1}^T \frac{\partial \ell_n}{\partial X_n} \frac{\partial X_n}{\partial \theta}, \quad (13)$$

where $\ell_n = \|\eta(X_n; \xi_n)\|_2^2/T + \lambda_u u_n^2/T$, and $\partial X_n/\partial \theta$ is itself obtained recursively from

$$\frac{\partial X_{n+1}}{\partial \theta} = \frac{\partial \Phi_{\Delta t}}{\partial X_n} \frac{\partial X_n}{\partial \theta} + \frac{\partial \Phi_{\Delta t}}{\partial u_n} \frac{\partial u_n}{\partial \tilde{u}_n} \frac{\partial \tilde{u}_n}{\partial \theta}.$$

The important term is $\partial \Phi_{\Delta t}/\partial u_n$: changing the neural control changes the next population profile, which changes future η , which changes the loss. Modern autodiff computes the same chain rule directly from the PyTorch tensor graph; no analytic reference controller is needed.

Question. Do we differentiate through the Lotka–Sharpe root solve?

In the first implementation, no. The hidden ecological profile ξ_n , including ζ_n , $\pi_{0,n}$, κ_n , and the instantaneous equilibrium profile x_n^* , is treated as environment truth at step n . These quantities define the measurement $\eta(X_n; \xi_n)$, but their root-solve construction is detached from the computational graph. Gradients are still taken with respect to the population state X_n , the control u_n , the controller hidden state, the self-talk message, and the network parameters θ .

Question. What happens at continuous-window boundaries?

Within a window, the graph contains all steps

$$(X_0, m_{-1}) \rightarrow (u_0, X_1, m_0) \rightarrow \dots \rightarrow (u_{T-1}, X_T, m_{T-1}),$$

so gradients flow through both the PDE state and the self-talk state. At the next optimizer update, the numeric values of X_T and m_{T-1} are carried forward, but the graph is detached:

$$X_T \leftarrow \text{stopgrad}(X_T), \quad m_{T-1} \leftarrow \text{stopgrad}(m_{T-1}).$$

This preserves the one-way ecological time stream while keeping each optimization problem finite and memory-bounded.

7 Differentiable PDE Tooling

The first implementation should use a PyTorch differentiable age-grid solver. Let

$$a_j = j\Delta a, \quad j = 0, \dots, N_a.$$

Use a first-order upwind update for the transport term:

$$\frac{x_{i,j}^{n+1} - x_{i,j}^n}{\Delta t} + \frac{x_{i,j}^n - x_{i,j-1}^n}{\Delta a} = -x_{i,j}^n \mathcal{R}_{i,j}^n, \quad j \geq 1, \quad (14)$$

where $\mathcal{R}_{i,j}^n$ contains mortality, control, and interaction terms. The boundary is updated by quadrature:

$$x_{i,0}^{n+1} = \sum_{j=0}^{N_a} k_i(a_j, t_n) x_{i,j}^n \Delta a. \quad (15)$$

All state updates, quadratures, η computation, softplus control, and losses should remain in PyTorch tensors so that the equilibrium loss can backpropagate through the rollout to the controller.

8 Time-Varying Ecology

The ecology should not be static. The first version uses the same profile families used in the Lotka-Sharpe neural-operator experiments [1]:

$$k(a, t) = k_{\text{base}} + k_{\text{amp}}(t) \exp\left(-\frac{(a - k_{\text{center}})^2}{2k_{\sigma}^2}\right), \quad (16)$$

$$\mu(a, t) = \mu_{\text{base}}(t) + \mu_{\text{juv,amp}} e^{-\mu_{\text{juv}} a} + \mu_{\text{sen,amp}}(t) a^{\mu_{\text{sen}}}, \quad (17)$$

$$g(a) = g_{\text{base}} + g_{\text{amp}} \exp\left(-\frac{(a - g_{\text{center}})^2}{2g_{\sigma}^2}\right). \quad (18)$$

The centers, widths, and interaction profile are fixed in the first version. The ecological regime changes through mixed-sin modulation of amplitude-like parameters:

$$k_{\text{amp}}(t), \quad \mu_{\text{base}}(t), \quad \mu_{\text{sen,amp}}(t).$$

A generic mixed-sin modulation is

$$q(t) = q_0 \left[1 + \rho \frac{\sum_{\ell=1}^L A_{\ell} \sin(\omega_{\ell} t + \varphi_{\ell})}{\sum_{\ell=1}^L |A_{\ell}|} \right], \quad 0 < \rho < 1. \quad (19)$$

The model does not observe $q(t)$, $k(a, t)$, $\mu(a, t)$, or $\zeta(t)$. It only observes $\eta_1(t)$, $\eta_2(t)$, and the constant pulse. The language loop is therefore tested as an internal state that may help track hidden ecological regime changes.

9 Continuous Training and Testing

Training should follow the exp0522 continuous case:

- time moves forward in one long ecological stream;
- training windows are consecutive pieces of this stream;
- the PDE state, language state, and error view are carried forward;
- the carried states are detached at window boundaries;
- the stream is not rewound to $t = 0$ after each update.

Evaluation should reserve a future segment of the same ecological stream. The model trains on the earlier segment, then rolls into the held-out future segment without weight updates. This tests whether the learned closed-loop controller continues to stabilize the ecology as the hidden birth and mortality parameters keep evolving.

10 First Experiment Design

The first experiment should compare:

1. **full self-talk controller:** η , pulse, error view, and language all live;
2. **no-language controller:** same external observations, but $m_t \equiv 0$;
3. **no-error-view controller:** language live, but $e_t \equiv 0$;
4. **minimal controller:** no language and no error view, only $(\eta_1, \eta_2, 1)$;
5. **blink/stutter rollouts:** temporary removal of error or language during evaluation.

The primary metrics are:

- mean and final $\eta_1^2 + \eta_2^2$;
- mean control energy u^2 ;
- positivity of $x_i(a, t)$ and $u(t)$;
- future-segment stability after training stops;
- recovery after blink/stutter interruptions.

11 Working Summary

The paper supplies a biologically meaningful full PDE environment and equilibrium coordinates. Exp0522 supplies the self-talk MLP and continuous rollout protocol. Exp0526 combines them by training a controller directly from closed-loop equilibrium loss:

$$(\eta_1(t), \eta_2(t), 1; \text{self-talk}) \mapsto \tilde{u}_t \mapsto u_t = \text{softplus}(\tilde{u}_t) \mapsto x(\cdot, t+1) \mapsto \eta(t+1) \mapsto \mathcal{L}_\eta.$$

References

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